

NurseFlow — Full Platform Documentation

Enterprise Healthcare Scheduling, Gemma 4 AI Patient Simulator & Break Lounge | Complete System Reference

1. Executive Overview & Platform Vision

NurseFlow is an enterprise-grade full-stack web platform designed to solve modern hospital workforce challenges, nurse fatigue, and clinical communication gaps. It combines automated smart roster scheduling, dynamic shift-swap workflows, Gemma 4 AI patient roleplay simulation, and a 10-minute shift break relaxation lounge into a unified glassmorphic interface.

Core Objectives:

- **Eliminate Duty Conflicts:** Standardized shift timings (Morning, Evening, Night) with clear weekly/monthly card views.
- **Elevate Clinical Communication:** Practice high-stress patient/family conversations using Gemma 4 AI with 8-competency evaluations.
- **Reduce Shift Burnout:** Non-medical break lounge mini-games (Tick & Cross) to revitalize focus during 10-minute shift breaks.
- **Instant Alert Broadcasts:** Socket.IO WebSocket push notifications for emergency ward calls and swap approvals.

2. Technical Architecture & Technology Stack

Component	Technologies Used	Purpose
Frontend Framework	React 18 + Vite, TypeScript	High-performance client UI rendering
Design System	Tailwind CSS, Glassmorphism, Lucide Icons	Modern aesthetics & responsive UI
Backend Server	Node.js, Express, REST API	API middleware, Auth & Routing
Database & ORM	Prisma ORM, SQLite (dev.db)	Data persistence & schema migrations
AI Intelligence	Gemma 4 AI Engine, Minimax Algorithm	Roleplay dialogue & Tic-Tac-Toe AI
Real-Time Engine	Socket.IO WebSockets	Instant push notifications & alerts

3. Core Module 1: Duty Roster Calendar & Schedule Management

The Roster Calendar module provides nurses and administrators with a spacious, color-coded weekly and monthly duty schedule. Cards are formatted to prevent text overlapping regardless of viewport width.

Standardized Shift Definitions:

- **Morning Shift:** 7:00 AM – 3:00 PM (Soft Emerald Tint #ECFDF5)
- **Evening Shift:** 3:00 PM – 11:00 PM (Soft Amber Tint #FFFBEB)
- **Night Shift:** 11:00 PM – 7:00 AM (Soft Purple Tint #EDE9FE)
- **Off Duty:** No Duty Assigned (Soft Slate Tint #F8FAFC)

4. Core Module 2: Gemma 4 AI Patient Communication Simulator

The AI Patient Communication Simulator allows nurses to practice healthcare communication in simulated scenarios with realistic patient and caregiver personas. It is an educational tool designed to build empathy and de-escalation skills.

Key Features:

- **Pre-Seeded Scenarios:** Anxious Patient, Angry Family Member, Confused Elderly Patient, Non-Cooperative Patient, Difficult Conversation, Emergency Triage.
- **Dynamic Character Emotion Tracking:** The character's emotional state transitions in real time based on nurse empathy signals (Hostile → Frustrated → Worried → Calm → Reassured).
- **8-Competency Evaluation Report:** Post-simulation analysis scoring Empathy, Active Listening, Clarity, Professionalism, Emotional Intelligence, De-escalation, Patient Engagement, and Confidence (0–100).

5. Core Module 3: Gemma 4 AI Tick & Cross (Tic-Tac-Toe ■■■) Break Game

Designed specifically for 10-minute shift breaks, this 100% non-medical relaxation game allows nurses to play Tic-Tac-Toe against Gemma 4 AI to unwind and reset focus.

- **3 AI Difficulty Modes:** Easy (Casual Chill), Medium (Smart Rival), and Unbeatable (Gemma 4 Minimax Algorithm).
- **Scoreboard & Wellness Points:** Tracks Nurse Wins, AI Wins, Draws, and awards Wellness Points.
- **10-Minute Break Timer:** A live countdown timer keeps break durations structured.

6. Backend REST API Endpoints Reference

Method	Endpoint	Description	Access
POST	/api/auth/login	User login & returns JWT token	Public
POST	/api/auth/register	Create new staff account	Public
GET	/api/rosters	Fetch weekly duty schedules	Auth
POST	/api/rosters	Assign roster to nurse	Admin
POST	/api/leave-requests	Submit leave application	Nurse

POST	/api/shift-swaps	Request duty exchange with peer	Nurse
POST	/api/notifications/send-alert	Broadcast ward emergency alert	Admin
GET	/api/communication-simulator/scenarios	List roleplay scenarios	Auth
POST	/api/communication-simulator/sessions/:id/messages	Send dialogue & get AI roleplay reply	Auth
POST	/api/communication-simulator/sessions/:id/end	End simulation & get 8-score report	Auth
POST	/api/games/tic-tac-toe/move	Calculate Gemma 4 AI move	Auth
POST	/api/games/tic-tac-toe/score	Save match outcome & wellness points	Auth

7. Database Schema & Models Data Dictionary

- **User:** id, name, email, passwordHash, role (NURSE, ADMIN, HEAD_NURSE), employeeId, departmentId, avatar.
- **Roster:** id, nurseId, shiftId, departmentId, date, status (ON_DUTY, SCHEDULED, OFF), notes.
- **CommunicationScenario:** id, title, category, description, characterRole, personality, difficulty, objectives.
- **CommunicationSession & Analysis:** id, userId, scenarioId, status, overallScore, empathyScore, activeListeningScore, clarityScore, strengths, feedback.
- **NurseGameScore:** id, userId, gameType, nurseWins, aiWins, draws, pointsEarned.